

LITERATURE SURVEY

IPS18 — Automated Healthcare Document Processing and Intelligence Platform

Track A — Artificial Intelligence & Machine Learning
Department of Computer Science & Engineering
B.E. Final Year Capstone Project — Semester 7

1. Introduction

This literature survey maps prior published work — primarily from IEEE Xplore — against each stage of the proposed IPS18 pipeline: OCR-based document ingestion, OCR-noise handling in medical documents, clinical named entity recognition, document classification, and RAG-based clinical search. The survey establishes that each individual pipeline component is independently well-supported in the literature, while identifying the specific combination — noise-adapted named entity extraction evaluated on OCR-degraded clinical documents — as the gap this project addresses.

2. Survey by Pipeline Stage

2.1 OCR / Scanned Document Text Extraction

Paper	Key Finding	IEEE Xplore Link
An Algorithmic Approach for Text Recognition from Printed/Typed Text Images	OCR using Otsu segmentation and Hough transform for skew detection; overall system accuracy of 93% on typed/scanned text.	ieeexplore.ieee.org/document/8442875
Analysis of Image Classification for Text Extraction from Bills and Invoices	OCR-based structured field extraction from scanned bills and invoices.	ieeexplore.ieee.org/document/9225564
An approach for Text Recognition from Document Images	General OCR-based text recognition pipeline for scanned document images.	ieeexplore.ieee.org/document/9297939
Text Extraction from Image Using OCR	Tesseract-based OCR with text localization, segmentation, and binarization for editable/searchable output.	ieeexplore.ieee.org/document/10883061
Optical Character Recognition (OCR) for Text Recognition and its Post-Processing Method: A Literature Review	Confirms OCR has not achieved 100% accuracy despite decades of development, motivating post-processing correction methods.	ieeexplore.ieee.org/document/9935961
Automated Text Extraction from Images using OCR System	Tesseract-based pipeline covering text detection, localization, segmentation, and binarization.	ieeexplore.ieee.org/document/8991281
Small Text Extraction from Documents and Chart Images	Deep learning (CRNN with CTC loss) improves small/degraded text recognition; super-resolution pre-	ieeexplore.ieee.org/document/10039990

Paper	Key Finding	IEEE Xplore Link
	processing improves accuracy by 18%.	

Table 2.1 — OCR / Scanned Document Text Extraction

2.2 OCR Noise & Correction in Medical/Clinical Documents

This category contains the papers most directly relevant to the project's core novelty — handling OCR-induced noise specifically in medical/clinical documents.

Paper	Key Finding	IEEE Xplore Link
An OCR Post-Correction Approach Using Deep Learning for Processing Medical Reports	Reports 2,314 exabytes of medical data generated globally in 2020, mostly unstructured scanned documents; OCR accuracy varies significantly due to inability to correctly transcribe skewed, obscured, or illegible scanned/handwritten medical text. Most directly relevant paper to this project's core problem.	ieeexplore.ieee.org/document/9448197
Customised OCR correction for historical medical text	OCR-recognised text from scanned medical page images often contains a large number of errors, since OCR systems are typically optimised for cleaner, non-domain-specific text.	ieeexplore.ieee.org/document/7413829

Table 2.2 — OCR Noise & Correction in Medical/Clinical Documents

2.3 Clinical NLP / Named Entity Recognition

Paper	Key Finding	IEEE Xplore Link
Clinical Text Analysis with Natural Language Processing: A BERT-based Approach	BERT-based approach spanning NER, clinical content classification, and topic modelling; reports NER precision/recall of 0.92/0.88 on clean clinical text.	ieeexplore.ieee.org/document/10593504
Extraction of Unstructured Electronic Healthcare Records using NLP	NLP-based extraction via classification, segmentation, and word embeddings; identifies data sensitivity and limited data availability as key challenges.	ieeexplore.ieee.org/document/10127351
Improving Clinical Report Classification with Sentence Boundary Detection	Benchmarks sentence-boundary detection methods to address LLM token-limit constraints on long clinical reports.	ieeexplore.ieee.org/document/11081534
Deep Learning Techniques on Text Classification Using NLP In Social Healthcare Network: A Comprehensive Survey	Survey of CNN/RNN deep learning techniques applied to healthcare text classification tasks.	ieeexplore.ieee.org/document/9451752
Applying Natural Language Processing in Healthcare Using Data Science	Review of NLP integration into clinical informatics research and healthcare workflows.	ieeexplore.ieee.org/document/10522196

Table 2.3 — Clinical NLP / Named Entity Recognition

2.4 Document Classification (Medical)

Paper	Key Finding	IEEE Xplore Link
Medical prescription classification: a	OCR followed by NLP/ML classification of scanned	ieeexplore.ieee.org/

Paper	Key Finding	IEEE Xplore Link
NLP-based approach	medical prescriptions by patient, doctor, symptom, and diagnosis fields.	document/8860013
Medical Document Classification Based on MeSH	NLP-based classification of medical journal articles against Medical Subject Heading (MeSH) categories.	ieeexplore.ieee.org/document/6758924

Table 2.4 — Document Classification (Medical)

2.5 Clinical Search / RAG-Based Question Answering

Paper	Key Finding	IEEE Xplore Link
Medical Information Extraction With NLP-Powered QABots: A Real-World Scenario	Semi-automated end-to-end pipeline (NEMT) deployed across a 16-hospital Italian consortium, centred on a QA-Bot extracting information into a centralized database.	ieeexplore.ieee.org/document/10651607
Efficient and Reproducible Biomedical Question Answering Using Retrieval Augmented Generation	Evaluates RAG retrieval strategies (BM25, BioBERT, MedCPT) and data stores (FAISS, Elasticsearch, MongoDB) at PubMed scale (24M documents).	ieeexplore.ieee.org/document/11081505
Medical Multi-Choice Question Answering with Retrieval Augmented Language Models	Notes that pre-trained language models lacking medical knowledge fail on professional medical questions without retrieval augmentation.	ieeexplore.ieee.org/document/10832057
Implementation of RAG Based Question-Answering Application	RAG-based chatbot implemented using LangChain, Chroma, and an LLM — directly aligned with the proposed IPS18 technology stack.	ieeexplore.ieee.org/iel8/11031246/11031251/11031968.pdf
RAG Powered LLMs for QA: Evolution, Challenges, Applications, and Future Directions	Comprehensive review of RAG addressing LLM hallucination, outdated knowledge, and untraceable reasoning via external retrieval.	ieeexplore.ieee.org/iel8/11033710/11034386/11034531.pdf

Table 2.5 — Clinical Search / RAG-Based Question Answering

3. Identified Research Gap

Sections 2.1, 2.3, 2.4, and 2.5 collectively demonstrate that OCR-based text extraction, clinical NER, medical document classification, and RAG-based clinical question answering are each independently well-established research areas with mature IEEE-published methods. However, this maturity is consistently demonstrated on clean, well-scanned, or typed text.

Section 2.2 confirms that OCR accuracy on real scanned medical documents is measurably compromised by skew, blur, and illegibility, and that OCR-recognised text in the medical domain retains a large number of errors even after standard correction. Despite this documented degradation, no surveyed work quantifies its downstream impact specifically on clinical named entity extraction, nor evaluates a noise-adapted fine-tuning strategy for this task in the clinical domain.

This project's contribution is therefore positioned precisely at this intersection: constructing an original OCR-degraded clinical NER dataset, fine-tuning a clinical language model against it, and honestly quantifying the accuracy gap between clean-text and noisy-OCR extraction — a combination not addressed by the papers surveyed above.

4. References (IEEE Format)

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